

Best Practices for the Synthesis, Activation, and Characterization of Metal–Organic Frameworks

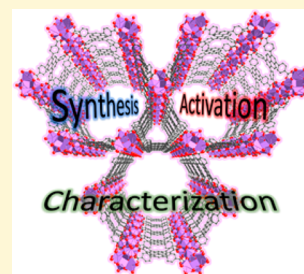
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S Supporting Information

ABSTRACT: Metal–organic frameworks (MOFs) are structurally diverse materials comprised of inorganic and organic components. As the rapidly expanding field of MOF research has demonstrated, these materials are being explored for a wide variety of potential applications. In this tutorial review, we give an overview of the current best practices associated with the synthesis, activation, and characterization of MOFs. Methods described include supercritical CO₂ activation, single crystal X-ray diffraction (XRD), powder X-ray diffraction (PXRD), nitrogen adsorption/desorption isotherms, surface area calculations, aqueous stability tests, scanning electron microscopy (SEM), inductively coupled plasma optical emission spectroscopy (ICP-OES), nuclear magnetic resonance spectroscopy (NMR), and diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS). A variety of different MOFs are presented to aid in the discussion of relevant techniques. In addition, some sections are accompanied by instructional videos to give further insight into the techniques, including tips, tricks, and suggestions only those at the bench could describe.



INTRODUCTION

In 1965, Tomic reported coordination polymers constructed using di- and tetraprotic carboxylic acid linkers coordinated to di-, tri-, or tetravalent metals.¹ The thermal stabilities of the polymers were studied and correlated to the valency of the metal used as well as the number of possible coordination sites on the linker.¹ The emergent properties of the materials that would later be known as metal–organic frameworks (MOFs) was just beginning to be understood. Nearly 25 years later, Hoskins and Robson proposed that a wide range of scaffold-like materials with infinite 3D frameworks should be accessible, tunable, and potentially useful.² Hoskins and Robson predicted that materials with large empty cavities and low densities should be accessible while maintaining high thermal, chemical, and mechanical stability. Some applications in molecular sieving, ion exchange, and catalysis were even suggested. A few years later, Yaghi et al. reported the use of hydrothermal synthesis to obtain a 3D crystalline and open material and coined the term metal–organic framework (MOF).³ Around the same time, the use of MOFs for methane gas storage was just beginning to be explored by Kitagawa et al.⁴ In 1999, Yaghi et al. reported MOF-5, the first framework to demonstrate permanent porosity and avoid structural collapse when guest solvent molecules were removed from its pores.⁵ Not long after that, the use of computational predictions and rational design by Férey et al. led to the synthesis of a highly stable MOF with very large pores (30–34 Å) and high surface area.⁶ Today there are thousands of MOF structures in the Cambridge Structural Database including frameworks with densities as low as 0.13 g/cm³,⁷ pore volumes up to 90% free volume,⁸ and Brunauer–Emmett–Teller (BET) areas greater than 6000 m²/g.^{8–11}

Given that MOFs are constructed from a variety of inorganic nodes (i.e., metal clusters or ions) and organic linkers, the number of metal–organic combinations, and therefore structural possibilities, are nearly endless. The highly modular and tunable nature of MOFs has fueled the study of these materials for a wide variety of potential applications including, but not limited to, gas storage and release,^{12–16} chemical separations,^{17–21} catalysis,^{22–24} drug delivery,^{25,26} light harvesting and energy conversion,^{27–29} sensing,^{30,31} conductivity,³² ion-exchange,³³ removal of toxic substances from air and water,^{34,35} and the degradation of chemical warfare agents.^{36,37}

Here, we describe the current state-of-the-art methods for the synthesis, activation, and characterization of MOFs. This overview is intended to help scientists who are new to the field of MOF research to become acquainted with the various techniques utilized. Finally, for the seasoned MOF researcher, this tutorial can initiate a discussion regarding the current best practices in the field.

Given that the field of MOF research is incredibly diverse, we will primarily cover the most common concepts and methods that are applied to synthesize, activate, and characterize these materials. Consequently, important, but more specialized, topics such as synthesis of hybrid MOF/organic–polymer, nanoparticle@MOF, and enzyme@MOF materials; characterization of MOF mechanical properties; and many other topics

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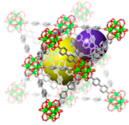
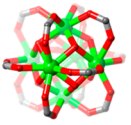
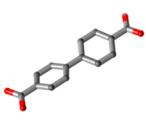
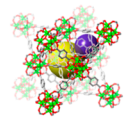
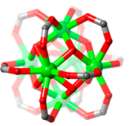
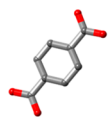
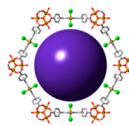
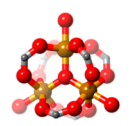
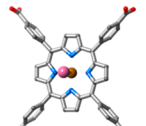
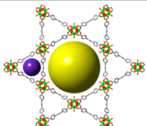
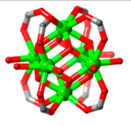
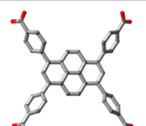
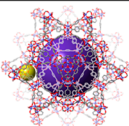
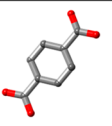
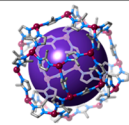
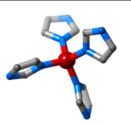
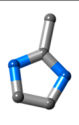
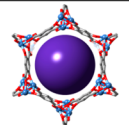
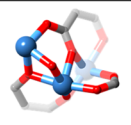
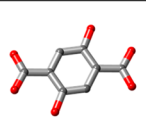
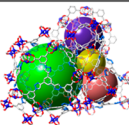
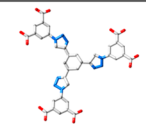
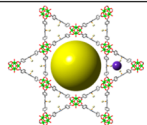
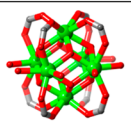
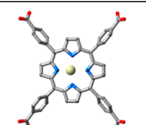
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will necessarily escape discussion. Table 1 shows the MOFs that will be discussed and outlines the metal node and organic linker components of each framework.

■ SYNTHESIS

A remarkably wide variety of methods can be used to synthesize MOFs.^{38–42} In general, the conditions for MOF synthesis should be chosen so that metal–ligand bonds can be formed

Table 1. Representations of MOF Structures and the Corresponding Node and Linker Constituents^a

MOF	Metal Node	Organic Linker
 UiO-67	 $\text{Zr}_6\text{O}_4(\text{OH})_4$	 BPDC
 UiO-66	 $\text{Zr}_6\text{O}_4(\text{OH})_4$	 BDC
 PCN-600(M)	 $\text{Fe}_3\text{O}(\text{OH})_3$	 M-TCPP
 NU-1000	 $\text{Zr}_6\text{O}_4(\text{OH})_6(\text{H}_2\text{O})_4$	 TBAPy
 Cr-MIL-101	 $\text{Cr}_3\text{O}(\text{OH})_3$	 BDC
 ZIF-8	 Zn	 MeIM
 Mg-MOF-74	 MgOH	 $\text{BDC}(\text{OH})_2$
 NU-125	 Cu_2	 LH ₆
 PCN-222(Fe)	 $\text{Zr}_6\text{O}_4(\text{OH})_6(\text{H}_2\text{O})_4$	 Fe-TCPP

^aZr: green; Fe: yellow; Cr: light purple; Zn: dark red; Mg: blue; Cu: royal blue; C: grey; O: red; N: light blue; Cl: pink.

but also broken and reformed to allow for structure propagation. Dynamic bonds are key to forming crystalline and ordered materials so that any erroneous bonding that may cause disorder or premature structure termination can be corrected. Solvothermal synthesis is the most straightforward and widely used method. Typically it entails mixing a metal salt with a multitopic organic linker in a high boiling point solvent (e.g., DMF, DEF, or DMSO) in a screw-top vial. The mixture is then heated, usually in an oven or on a hot plate sited in a fume hood and equipped with a bath of nonflammable silicone-based oil (*not* mineral oil), typically for 12 to 48 h. Parameters that can be systematically varied include reaction temperature, time, solvent, reagent concentration, pH, and nature of the precursors used. These parameters may affect not only the topology obtained but also the crystal size and phase purity of the material. In this conventional synthetic approach, the metal precursor and organic linker should at least be somewhat soluble when the mixture reaches the target temperature. It is also important to know the balanced reaction for MOF synthesis. By design, MOF synthesis is dynamic and can be very sensitive to small variations in the reaction mixture. A metal chloride (MCl_x) mixed with a multitopic carboxylic acid linker, for example, will generate at least a stoichiometric amount of HCl, a strong acid which may dissolve the MOF that is forming and slow crystal growth. Alternatively, using a metal acetylacetonate ($\text{M}(\text{acac})_x$) produces acetylacetone as a byproduct ($\text{pK}_a \approx 9$ in water) which is more mild and less likely to affect crystal growth. Lastly, when choosing the reaction vessel (typically a screw-top vial), the scale of the reaction and hence the volume of solvent needed as well as the target temperature should be considered to ensure that sufficient headspace is left to allow for potential pressure buildup.

In some instances—particularly when the metal–ligand bonds are very strong—a modulator can be used to help prevent rapid precipitation of amorphous material.^{43–46} Modulators are nonstructural, monotopic linkers (e.g., benzoic acid, acetic acid, hydrochloric acid), which can form dynamic bonds with the metal precursor and help to slow down the formation of structural bonds by competing with the linkers for metal coordination sites. The use of modulators has been particularly important for the synthesis of Zr-MOFs which contain strong Zr(IV)–O bonds. Video 1 shows the synthesis of a Zr-MOF, UiO-67 (Table 1), which is made using the conventional approach as described above. In Video 1, procedures depicting the use of both hydrochloric acid⁴⁷ and benzoic acid⁴³ as modulators are shown. Although the two kinds of modulators yield compounds featuring the same topology, with both correctly being designated UiO-67, the two compounds are not identical. For example, the procedure utilizing benzoic acid modulator typically results in a close to ideal structure, in which the Zr_6 nodes of the framework are connected to 12 structural linkers, whereas the procedure using hydrochloric acid as the modulator typically results in a defective structure where some structural linkers are missing, forming defects sites (i.e., nodes terminated by hydroxyl, water, or chloride ligands). A thorough discussion of defects in Zr-MOFs (or MOFs in general^{48,49}) is outside of the scope of the tutorial.^{44–46,50,51} It is important to appreciate that defects are not necessarily undesirable; they can favorably influence MOF surface areas, porosities, and, especially, catalytic activity. Returning to modulators, their chemical composition and synthesis concentration not only can affect defects but also can

have bearing on the MOF crystal size, habit, and topology obtained.^{52–55}

A variant of the conventional MOF synthesis involves the preformation of metal nodes (clusters) or secondary building units (SBUs). Férey et al. demonstrated that replacing ZrCl_4 with a Zr_6 -methacrylate cluster $[\text{Zr}_6\text{O}_4(\text{OH})_4(\text{OMc})_{12}]$ where OMc = methacrylate can be used as an alternative route to form UiO-66 (Table 1) and its derivatives.⁵⁶ A different example from Zhou et al. involves the synthesis of a series of metalloporphyrin MOFs with Fe_3 nodes, PCN-600(M) (Table 1), which were obtained by first synthesizing and isolating the metal cluster nodes $[\text{Fe}_3\text{O}(\text{OOCCH}_3)_6\text{OH}]$.⁵⁷ The cluster was then mixed with a tetratopic porphyrin-based linker and an acid modulator in DMF and subsequently heated in an oven overnight. Metal cluster nodes can also be generated in situ prior to the addition of structural organic linkers to the reaction mixture. Video 2 shows the synthesis of a Zr-MOF, NU-1000 (Table 1), which is made by first synthesizing the Zr_6 -oxide cluster (which is not isolated) followed by the addition of the organic linker.⁵⁸ By forming the metal clusters of NU-1000 first, the phase purity and surface area of the MOF is optimized. For more complex topologies, a bottom-up approach can be used where small molecular clusters are used to construct super(or supra)molecular building blocks (SBBs) or tertiary building units (TBUs) which can then be connected to form a 3D framework.^{59,60}

Alternative MOF synthetic strategies include electrochemical,^{61,62} mechanochemical,^{63–68} and sonochemical methods⁶⁹ as well as microwave assisted synthesis.^{69,70} For the synthesis of MOF thin films, layer-by-layer deposition, liquid phase epitaxial growth, or seeded growth on a coated substrate can be performed.⁷¹ Under-appreciated features of liquid-phase epitaxy are that mild conditions and very low linker concentrations (a few micromolar) can often be used. Thus, the methodology lends itself well to the assembly of MOFs from scarce, delicate, poorly soluble linkers and to the circumvention of unwanted thermally driven chemical transformations (e.g., metalation of porphyrinic linkers by ions intended for use only as nodes). For MOFs that are difficult to synthesize *de novo*, various postsynthetic methods have been developed including post-synthetic modification (PSM),⁷² solvent assisted linker exchange (SALE),⁷³ and transmetalation.^{74,75} In the latter two, the organic linkers or metal nodes in an MOF with a given topology can be replaced with new linkers or metals to obtain a new framework with the same parent topology.

■ ACTIVATION

During the synthesis of MOFs, solvent molecules are inevitably trapped in the pores of the framework. In a few notable cases, (e.g., Cr-MIL-101⁶), excess linkers may also be trapped. To access the permanent porosity and high surface areas promised by many framework structures, the solvent molecules must be removed—a process termed activation. Often, some care is needed when activating MOFs, if the goal is to access the highest possible surface area and porosity.⁶ The most common activation techniques will be discussed in the following sections.

Vacuum Drying and Solvent Exchange. In some instances, simply heating the MOF under vacuum directly after synthesis may suffice. This is the case for Cr-MIL-101⁶ and ZIF-8⁷⁷ (Table 1), which have BET surface areas of ~ 3500 m^2/g and 1700 m^2/g , respectively. In most cases, however, the direct application of heat and vacuum following MOF synthesis leads to lower surface areas than expected from *in silico*

assessments^{78,79} due to framework collapse. Framework collapse can often be attributed to the high surface tension and capillary forces imposed on the structure by the liquid-to-gas-phase transformation of trapped solvent molecules—especially when the solvent has a high boiling point and/or high surface tension.⁸⁰ The easiest way to overcome this problem is to exchange the solvent with a lower-boiling-point/lower-surface-tension solvent (these parameters tend to be correlated) prior to heating the sample under vacuum. Figure 1

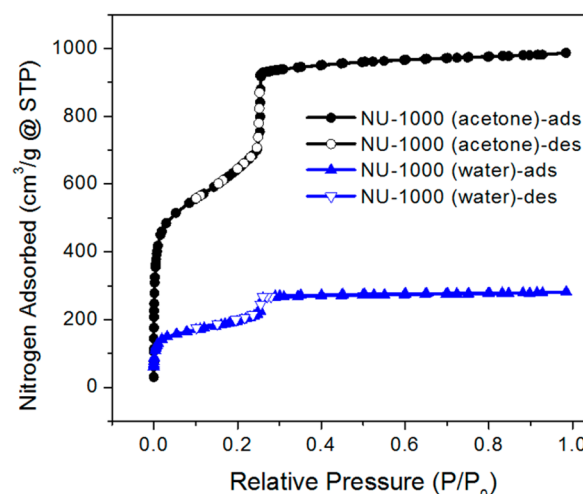


Figure 1. Nitrogen adsorption and desorption isotherms for NU-1000 activated from acetone versus water.

shows nitrogen adsorption/desorption isotherms for NU-1000 activated directly from water (framework collapse) compared to a sample activated from acetone. Video 3 presents some tips on how to properly perform a solvent exchange. First, the MOF should be washed thoroughly with the reaction solvent to remove any noncoordinated linkers or other impurities, which may also result in lower than expected surface area measurements. It is important throughout all washing steps (normally performed in a centrifuge tube) that the MOF is fully suspended prior to centrifugation and removal of the solvent. When exchanging with a lower boiling point solvent such as ethanol or acetone, the MOF should be left to soak in the new solvent, between washes, to ensure that the new solvent infiltrates the pores. This may also require soaking overnight or for a prolonged period of time (in some cases days) to ensure complete solvent exchange occurs. NMR spectroscopy can be used to confirm successful solvent exchange (see section below on NMR sample preparation). Once the solvent exchange is complete, heat and vacuum can be applied to complete activation of the sample. The activation temperature used should be above the boiling point of the solvent under vacuum and well below the decomposition temperature of the framework. Solvent exchange can also be performed by Soxhlet extraction if the reaction solvent is difficult to remove from the pores or if the linker used is not very soluble and difficult to separate—such is the case with Mg-MOF-74 (Table 1) and its derivatives.⁸¹

Supercritical Drying. An extension of conventional solvent exchange for MOF activation is exchange using supercritical carbon dioxide (scCO_2).^{76,82} scCO_2 activation is a more mild activation technique which may be required if conventional solvent exchange is unsuccessful and causes framework collapse.

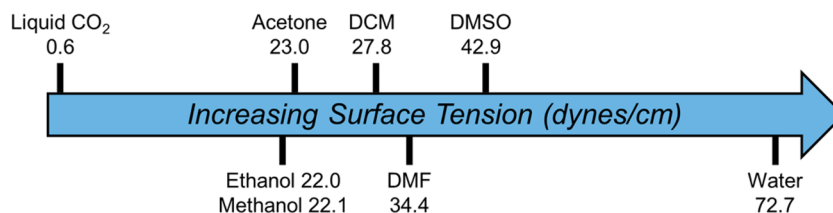


Figure 2. Surface tension of some common organic solvents⁸⁴ compared to liquid CO₂.⁸⁵ DCM: dichloromethane; DMF: *N,N*-dimethylformamide; DMSO: dimethyl sulfoxide.

scCO₂ activation is milder since it avoids the liquid- to gas-phase transformation of guest solvent and instead goes through a supercritical phase which eliminates surface tension and, therefore, capillary forces.⁷⁶ Figure 2 shows the surface tension of some common organic solvents compared to liquid CO₂. To prepare an MOF sample for scCO₂ activation, the sample is first subjected to conventional solvent exchange with a solvent that is miscible with liquid CO₂ (e.g., ethanol, methanol). The compatibility of the solvent with the specific scCO₂ dryer should also be confirmed prior to solvent exchange. Once the sample has been washed with the appropriate solvent, it should be soaked in the solvent overnight in a scCO₂ drying dish, and the sample should be allowed to settle to the bottom of the dish. Video 4 shows the important steps of scCO₂ activation using a commercially available and inexpensive instrument. After soaking, the solvent should be carefully removed from the sample using a pipet, leaving only a thin layer of solvent on top of the MOF. It is important that the MOF remains solvated throughout the process as allowing the sample to “dry” can cause framework collapse. The sample can then be placed in a scCO₂ dryer and cooled to 2–10 °C, and the solvent can be exchanged with liquid CO₂. Care should be taken to ensure that the temperature of the sample chamber does not drop below 0 °C to avoid possible issues associated with freezing trace water in the solvent, which can also impart strain on the framework. The liquid CO₂ should be purged from the sample every 1–2 h and exchanged with fresh liquid CO₂. After 3–4 exchange cycles, the sample can then be heated to the supercritical temperature and pressure of CO₂ (i.e., 31 °C and 73 atm) and the gaseous CO₂ slowly released from the sample (usually called “bleeding”) at a rate of 0.1–1 cm³/min. Prior to gas adsorption analysis, the sample should be placed under mild heat (<50 °C) and vacuum to remove any potentially physisorbed CO₂ leftover from the activation process. Figure 3 shows nitrogen adsorption/desorption isotherms for the Cu-MOF, NU-125 (Table 1), activated by solvent exchange compared to scCO₂ drying. In this example, the MOF does not completely collapse,⁹ but the highest possible surface area cannot be obtained by solvent exchange. It should be noted that in some cases scCO₂ activation can be performed by flowing scCO₂ over MOF crystals solvated with DMF.⁸³ The flow method may be faster and more scalable than conventional scCO₂ drying but requires a specialized and more expensive setup.⁸³

Given that adjustments to activation protocols can often yield higher surface areas, how does one know when to cease trying to improve an activation protocol? Comparisons of experimentally measured surface areas to calculated (i.e., computationally modeled) maximum achievable surface areas are often insightful. If an experimental apparent surface area (see below) approaches within a few to several percent of the computationally derived, maximum attainable surface area, the

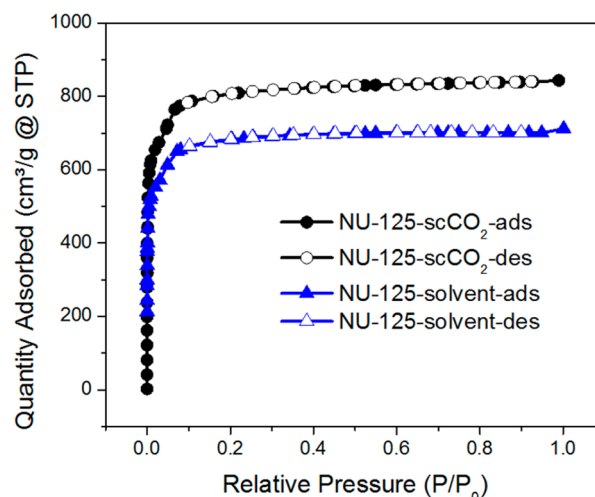


Figure 3. Nitrogen adsorption-desorption isotherm of NU-125 activated using solvent exchange compared to scCO₂ drying.

activation procedure can reasonably be regarded as well-optimized. Occasionally, measured apparent surface areas are found to substantially exceed theoretical maximum areas. If measurement errors and sample contamination can be correctly ruled out, the persistence of disparities in surface area assessments may be indicative of the existence of a preponderance of missing-node and/or missing-linker type defects. Methods such as DRIFTS (discussed below) and acid/base titration⁸⁶ can then be enlisted to characterize and quantify defects.

■ CHARACTERIZATION

Basic MOF characterization data include a powder X-ray diffraction (PXRD) pattern to establish crystallinity and phase purity of the material in addition to nitrogen (N₂) adsorption/desorption isotherms to confirm porosity and calculate an apparent surface area. Additional characterization techniques and protocols may include (i) thermogravimetric analysis (TGA) to determine the thermal stability of an MOF and in some cases to estimate the pore volume; (ii) aqueous stability testing to determine the stability of an MOF in water and at varying pH; (iii) scanning electron microscopy (SEM) to measure crystal size and morphology which can be coupled with energy dispersive X-ray spectroscopy (EDS) to learn more about elemental composition and distribution; (iv) NMR spectroscopy which can be used to determine the bulk purity of a sample in addition to quantifying linker ratios in mixed linker MOFs;^{87,88} (v) inductively coupled plasma optical emission spectroscopy (ICP-OES) which can be used to determine the purity of a sample as well as elemental ratios; (vi) diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) which can be used to confirm the presence (or absence) of IR

active functional groups in the framework; and finally (vii) single crystal X-ray diffraction (XRD) which gives absolute structural information. It is important that all characterization methods are used for the correct purpose—for example, PXRD patterns should not be used to confirm porosity of a material but instead should be used together with N_2 isotherms to understand crystallinity and porosity, respectively. Each of these characterization methods, focusing on sample preparation and data analysis, will be discussed briefly below.

Powder X-ray Diffraction. PXRD patterns are used to determine bulk crystallinity of MOF samples. Once a sample is determined to be crystalline, other information can be extracted from the powder pattern such as unit cell size. MOF phase purity can be confirmed by comparing the experimental powder pattern to simulated patterns generated from single crystal X-ray data or through the use of computational modeling.⁷⁹ One common method used for preparing samples for PXRD analysis involves loading the powdered sample onto a flat plate sample holder which can be made from plastic, glass, or aluminum. Samples can be dry loaded onto the holder or affixed using oil or volatile solvents. Although this sample preparation method works well for the majority of samples, it is important to remember that some crystals (especially those with plate or needle morphology) will deposit with a preferred orientation when prepared in this fashion. Given that simulated powder patterns are generally calculated based on the assumption that crystallites are randomly oriented, significant differences in peak intensities may exist between simulated versus experimental patterns. (Figure 4 shows one example). To avoid issues

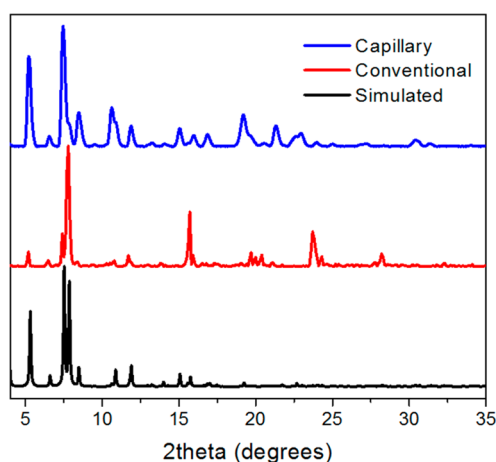


Figure 4. Powder X-ray diffraction pattern for MOF crystals with plate morphology run using a spinning capillary versus a conventional sample holder without spinning and compared to a simulated powder pattern. MOF is ZnZn-RPM.⁸⁹

associated with preferred orientation, the MOF sample should be continually rotated during data collection (in a conventional sample holder or in a capillary tube).⁸⁹ By spinning the sample, the orientation of the crystallites becomes nearly random in relation to the detector. Overly broad diffraction peaks typically are associated with undersized crystallites and concomitant Scherrer broadening, rather than with poor crystallinity. Full structural determination can also be accomplished using PXRD data; however, the process is challenging.^{90,91} Finally, the appearance of nominally symmetry forbidden, weak diffuse reflections can, in favorable cases, provide information about

crystal defects and the extent to which they are periodic (i.e., correlated⁹²).

Nitrogen Adsorption and Desorption Isotherms.

Adsorption isotherms for nonreactive gases at cryogenic temperatures can be used to determine apparent surface areas for MOFs, in addition to MOF pore volumes and pore size distributions. The shape of an adsorption/desorption isotherm can also give valuable information about the material.^{93,94} Nitrogen (N_2) gas adsorption at 77 K is most commonly employed for this purpose. To obtain meaningful information, the MOF should be properly activated (see [Activation](#) section above) before an isotherm is collected. The amount of sample used in the analysis is important, and as a rule-of-thumb the amount of sample (in grams) multiplied by the specific surface area (in m^2/g) of the sample should be equal to 100 m^2 or more to obtain reliable data. For the isotherms to yield reliable results, the choice of data analysis protocol is crucial. If the analysis is done incorrectly, it is remarkably easy to either underestimate or overestimate apparent surface areas.^{95–97} At present, the best default practice is to use BET theory⁹⁸ for apparent surface area calculations since the pore sizes of most MOFs support multilayer gas adsorption. The Langmuir theory is valid only for monolayer adsorption⁹⁹ and, as a result, tends to overestimate the apparent surface area of most MOFs—in some cases by 50% or more. The BET equation (eq 1) can be used to determine the apparent surface area of an MOF by plotting $[(p/p_0)/n(1 - p/p_0)]$ vs p/p_0 , where n is the amount of N_2 gas adsorbed, p is pressure, and p_0 is saturation pressure of N_2 . From this plot, a linear region must be selected based on criteria proposed by Rouquerol et al. which state that (i) the BET constant “C” must be positive, (ii) $n(1 - p/p_0)$ should increase monotonically with p/p_0 , (iii) the monolayer capacity (n_m) should correspond to a pressure within the limits of the data, and (iv) the calculated value for monolayer formation ($1/(\sqrt{C} + 1)$) should be approximately equal to p/p_0 at the monolayer capacity.¹⁰⁰ From the slope and intercept of the linear region, the BET area (apparent surface area), n_m and C can be extracted. Figure 5 and Table 2 show an example of how the region of the isotherm selected to calculate BET area can have a significant effect on the value obtained. Figure 5a shows a Rouquerol plot where only points to the left of the dotted line should be selected to meet criteria (ii) above. Figure 5c,d shows BET plots corresponding to different regions of the isotherm in Figure 5b where nonlinear (incorrect) and linear (correct) BET regions are chosen, respectively. As shown in Table 2, the correct choice of data points gives a BET area of 2200 m^2/g while an incorrect choice of points from the same isotherm gives a BET area of 4400 m^2/g . When two linear BET regions are available, Rouquerol criteria (iii) and (iv) above should be used to determine which linear region is the correct one to fit. Figure 5c,d also shows the incorrect and correct application of criteria (iii) and (iv), respectively. It should be noted that the data in Table 2 are a typical representation of values obtained when using software to fit N_2 isotherms. The actual reported values for BET area and other parameters should be rounded to the appropriate significant figures based on the accuracy of the instrument used and the precision with which the sample size is known. The empirical justification for the use of BET theory with MOFs has been explored in detail in the literature, in part by applying various forms of the analysis to computationally generated isotherms for perfect (in silico) MOF structures for which absolute surface areas can be independently calculated.^{95–97} Nevertheless, it is important to note that both BET

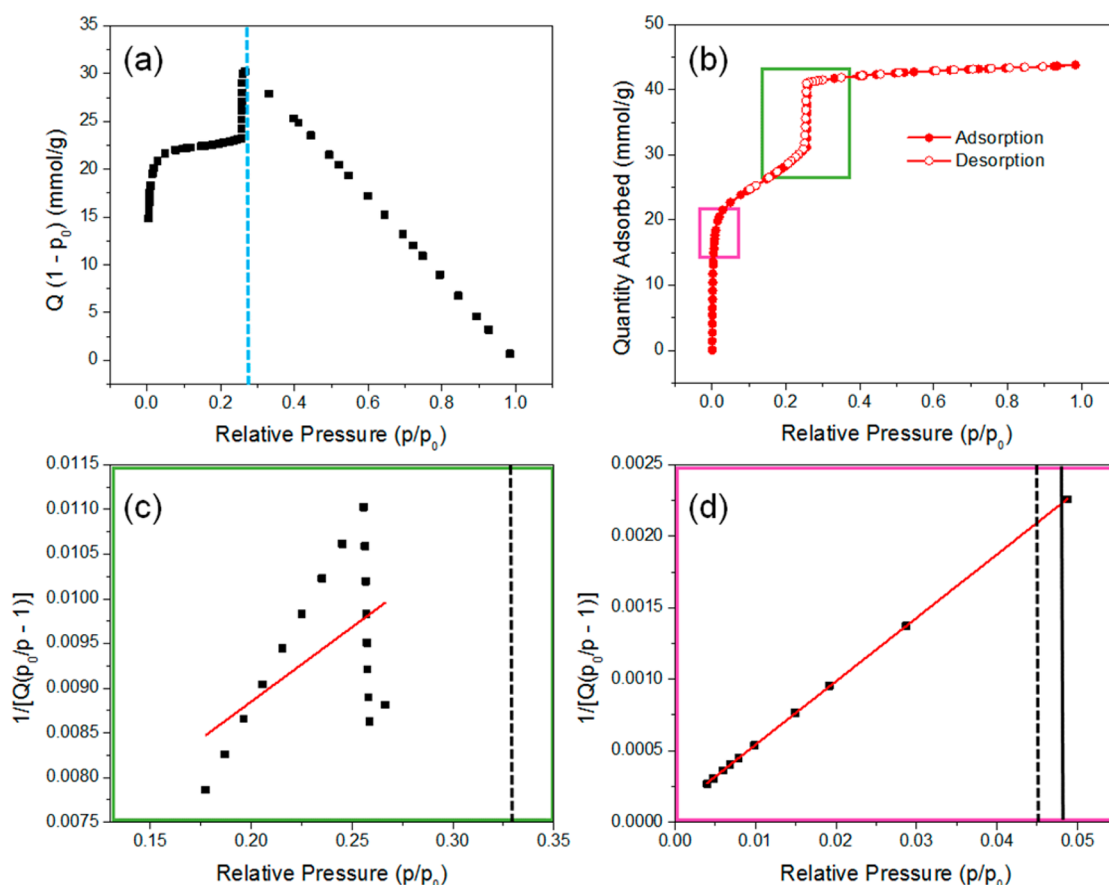


Figure 5. (a) Rouquerol plot where only points to the left of the dotted line should be selected; (b) nitrogen adsorption and desorption isotherm showing the regions selected for the BET plots in c (green) and d (pink); (c) BET plot taken from $p/p_0 = 0.17$ – 0.27 ; (d) BET plot taken from $p/p_0 = 0.004$ – 0.05 ; in c and d the solid line corresponds to p/p_0 at n_m (criterion (iii)) and the dotted line corresponds to $1/\sqrt{C + 1}$ (criterion (iv)); in (c) the solid line is not shown because the monolayer capacity is above the highest point on the isotherm in (b).

Table 2. Values Obtained for the BET Area, Slope, y-Intercept, C , n_m , R^2 , $1/\sqrt{C + 1}$, and p/p_0 at n_m Using the Fittings Shown in Parts c (green) and d (pink) of Figure 5

BET area	$4400.6556 \pm 1359.0268 \text{ m}^2/\text{g}$	$2195.0589 \pm 4.6446 \text{ m}^2/\text{g}$
Slope	$0.016647 \pm 0.006661 \text{ g}/\text{mmol}$	$0.044346 \pm 0.000094 \text{ g}/\text{mmol}$
Y-intercept	$0.005522 \pm 0.001583 \text{ g}/\text{mmol}$	$0.000099 \pm 0.000002 \text{ g}/\text{mmol}$
C	4.014631	449.779668
n_m	45.10758 mmol/g	22.49978 mmol/g
R^2	0.5422064	0.9999820
$1/\sqrt{C + 1}$	0.33	0.045
p/p_0 at n_m	N/A	0.049

and Langmuir theories were designed for modeling flat surfaces and therefore some assumptions are made when using either method with highly porous, 3D MOF materials. Perhaps MOF materials will drive the evolution of these models to more accurately describe the experimental data, and more MOF-specific models will be devised.

Similar to surface area modeling, it is important to be cautious when extracting information about the pore volume and pore size distribution of an MOF from an N_2 isotherm. There are multiple models that can be used to analyze pore volume and pore size distribution of an MOF including the density functional theory (DFT) method¹⁰¹ and Barrett–Joyner–Halenda method,¹⁰² among others. The DFT model is most widely accepted for the analysis of MOF pores—especially if mesopores close to 2 nm in width or micropores

(<2 nm) are present. Expected pore size distributions can be derived from X-ray crystal structures. Assuming that the appropriate model is used, and the sample being investigated is pure, the appearance of pores of unanticipated size may be a tip-off that defects in the form of missing (or excess) linkers or nodes are present in non-negligible quantity. Mismatches between anticipated and observed pore distributions, especially for MOFs featuring hierarchical porosity, can provide clues about pore-specific solvent occlusion or partial framework collapse. Finally, certain kinds of hysteretic behavior for adsorption versus desorption isotherms, especially with sorbates/guests other than N_2 and especially at elevated pressure, can provide information about guest-induced displacement of catenated rigid frameworks or guest-induced distortion of flexible frameworks, especially in the vicinity of apertures. The particulars of such behavior and analyses are beyond the scope of our article; however, it is important to appreciate that isotherm hysteresis can be caused by a range of phenomena—some of them chemically illuminating and some of them not.

$$\frac{p/p_0}{n(1 - p/p_0)} = \frac{1}{n_m C} + \frac{C - 1}{n_m C} (p/p_0) \quad (1)$$

Thermogravimetric Analysis. TGA is a commonly used technique to determine the thermal stability of an MOF and to estimate its solvent-accessible pore volume. When measuring the thermal stability of an MOF, the carrier gas species (e.g., N_2 , air, or O_2) chosen for the measurement is important and

should always be reported because the decomposition pathway of an MOF can vary in different environments (Figure 6 shows

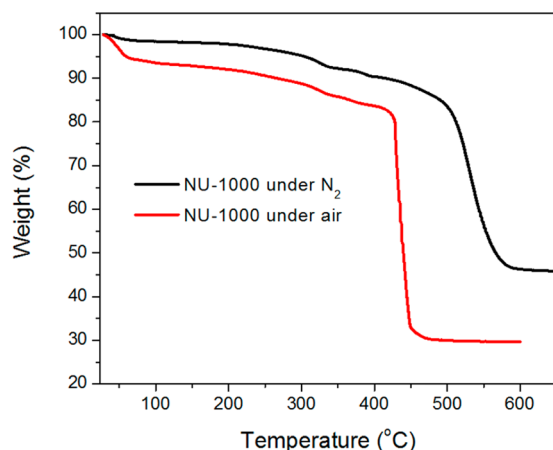


Figure 6. Thermogravimetric analysis curve showing weight % versus temperature taken under N_2 and air. MOF used is NU-1000 (Table 1).

an example). In addition, care should be taken when determining MOF thermal stability from TGA measurements as mass loss is not necessarily associated with structural change. Complementary measurements should always be taken to confirm stability of an MOF with varying temperature including in situ PXRD^{103,104} and/or sorption measurements performed on materials exposed to varying temperatures.¹⁰⁵

Pore volume can be roughly estimated by collecting the TGA spectrum of a solvated sample. In this case, mass loss attributed to trapped solvent in the pores can reflect the pore volume of the MOF sample. The MOF should first be soaked in a high boiling point solvent (e.g., DMF) for a sufficient amount of time to ensure diffusion of the solvent into the pores. The MOF can then be quickly filtered to remove any solvent

molecules that are not trapped in the pores. If performed correctly, the mass loss attributed to the release of the trapped solvent (near the solvent boiling point) should correspond to the void space of the MOF. It is important to remember, however, that the mass loss of the solvent can be larger than expected due to incomplete removal of the solvent in the interparticulate space or smaller than expected due to the removal of solvent in the pore when filtering the sample, so care should be taken when collecting and interpreting data using this method.

Aqueous Stability Testing. MOFs are often suggested as stable supports for catalysis and chemical storage. For some of these applications, the aqueous stability of the prepared materials is of significant importance. We suggest the following as a guide to test the aqueous stability of MOFs in a standardized fashion to allow for easier comparison among different materials. As many MOFs have functional groups which can be basic or acidic, it is important to use an adequate amount of water for each test (i.e., 20 mg of MOF suspended in 10 mL of water). The pH of the water should be tested before the addition of MOF and once the MOF is filtered and removed to ensure accurate results. To test the acid and base stability of an MOF, solutions with accurately measured pH (using a calibrated electronic pH probe) should be prepared using either HCl or NaOH, respectively. It is important that the chemical identities of the buffers (e.g., phosphates) and counterions (e.g., F^-) be considered when making pH solutions as these can interfere with the experiment and yield misleading results.^{106–108} For example, MOF degradation due to metal-ion extraction by strongly coordinating electrolyte components may be mistaken for degradation due to hydrolysis. Following solution preparation, an accurately weighed amount of the activated MOF should be suspended in an aqueous solution of the desired pH. This suspension should be left undisturbed for at least 12 h at room temperature. It is important to indicate the

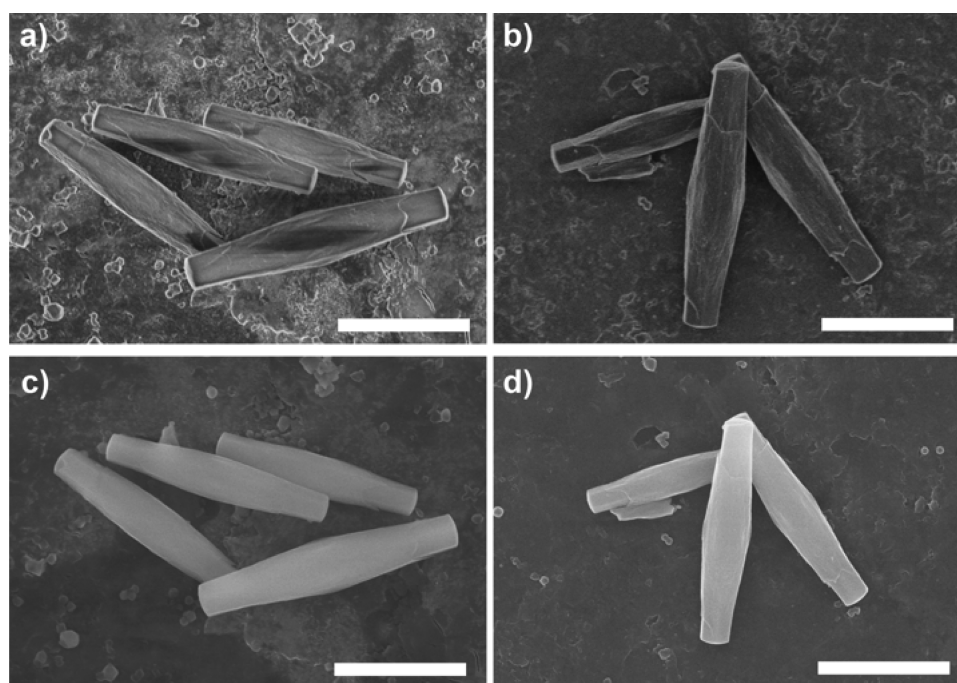


Figure 7. SEM images of PCN-222(Fe) with (a and c) and without (b and d) Os coating under 2 kV (a and b) or 15 kV (c and d) accelerating voltage. Scale bar represents 5 μm .

time and temperature of exposure to allow for accurate comparisons in future experiments. To access the stability of the MOF after exposure to the aqueous solution, the material should be recovered through careful filtration and reactivated following the appropriate activation procedure (see section on MOF activation above). The mass of material recovered should be noted, and by using eq 2, the percent of recovered material should be calculated to ensure that a portion of the MOF did not dissolve in the challenge solution. After mass balance has been determined, it is important that the identity of the material is verified by measuring the PXRD (see section above) and the N_2 isotherm (see section above) for comparison with the pristine material.

$$\frac{\text{mass}_{\text{recovered}}}{\text{mass}_{\text{initial}}} \times 100 = \text{yield } \% \quad (2)$$

Scanning Electron Microscopy. SEM is a useful tool to measure a variety of different MOF properties including crystal size, morphology, and elemental composition. If the size of the MOF crystals permits, optical microscopy can also be used to obtain basic information about crystal size and morphology. Due to the insulating nature of most MOFs, image artifacts, such as charging effects, can often detract, or in some cases completely obstruct, the acquisition of quality SEM images. The most common method of alleviating these issues is to coat the sample with a conducting material (e.g., gold or osmium) to decrease charge buildup from the electron gun. Figure 7a,b gives an illustrative example of an SEM image of PCN-222(Fe) (Table 1) with and without 5 nm of Os coating, respectively. It is directly apparent that charging effects have caused abnormal contrast in certain regions of the MOF crystallite, whereas there are no such apparent artifacts in the Os-coated sample. Additionally, selection of the appropriate accelerating voltage of the electron beam can dramatically change the appearance of the MOF crystals. Although greater image resolution is obtained at high accelerating voltages (Figure 7c,d), this comes at a loss of surface detail, prohibiting knowledge of surface defects or contamination. A larger accelerating voltage also increases the probability of local heating due to inelastic electron scattering which can induce structural damage to the MOF crystals.

SEM, coupled with EDS, can yield both quantitative and qualitative information regarding metal dispersion/incorporation in MOFs. Here, however, judicious selection of the coating material is necessary, as peaks from the coating, or even their contaminants (e.g., yttrium is often found in coatings with osmium) could overlap with metals of interest, introducing error in quantitative analysis (Figure 8). As an example, quantitative EDS measurements of Os-coated and Os-free crystals of PCN-222(Fe) yield Fe:Zr₆ ratios of 2.9 ± 0.5 and 2.0 ± 0.2 , respectively (an ideal PCN-222(Fe) crystallite should have an Fe:Zr₆ ratio of 2:1). Ideally, ICP-OES quantification should always be measured to confirm elemental composition.

Inductively Coupled Plasma Optical Emission Spectroscopy. ICP-OES or mass spectrometry (ICP-MS) can be used to confirm purity or elemental ratios in an MOF sample. For most elements, ICP-MS can be used to detect very low concentrations (as low as ppt) compared to ICP-OES (as low as ppb). The lower detection limits for ICP-MS may be important when the amount of sample to be analyzed is limited, but in general it is preferred to use more sample and work in higher concentration regimes to minimize error associated with sample preparation. For either analysis, the sample must first be

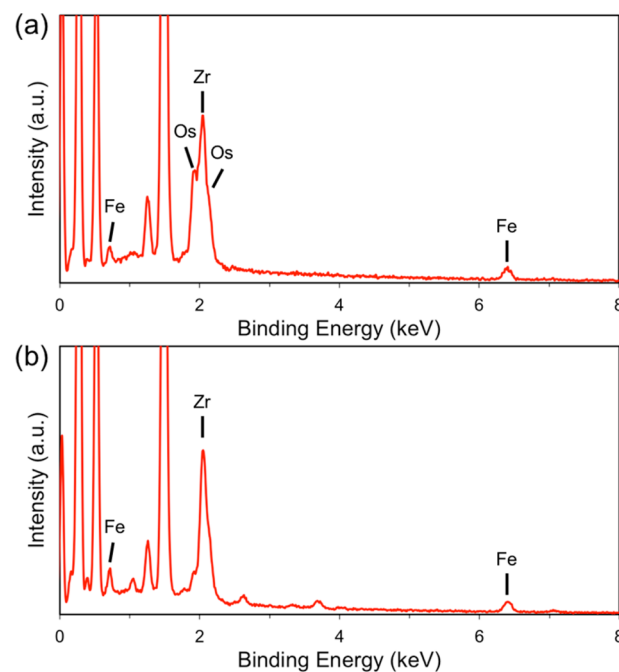


Figure 8. Energy dispersive spectra of a PCN-222(Fe) crystal (a) with and (b) without Os coating. Notice peaks from Os overlap with peaks from Zr.

completely digested (dissolved) before injection into the instrument. The most common matrices used for ICP-OES/MS samples include dilute (3–5%) acids such as nitric (HNO_3) or sulfuric (H_2SO_4) acid. Since many MOFs are not soluble in dilute acid at room temperature, different procedures for sample digestion should be used. Video 5 shows one possible procedure for MOF digestion prior to ICP analysis, which involves adding 750 μL of H_2SO_4 followed by 250 μL of hydrogen peroxide to 2–3 mg of an MOF in a microwave tube. The sample should then be heated using microwave irradiation at 150 $^{\circ}C$ for 5 min. After this time, the sample should be completely clear (not necessarily colorless), meaning that the MOF is fully digested before the sample is diluted with water (to give 3–5% acid concentration) for analysis. If the sample is not fully dissolved, the analysis will be meaningless and a different digestion procedure should be used. In the procedure described above, HNO_3 can be used in place of H_2SO_4 , especially if sulfur analysis of the MOF is required. If ICP-OES/MS is used to confirm purity, the sample must be carefully weighed out prior to digestion so that proper elemental weight percentages can be calculated and confirmed from the analysis. If all that is required is an elemental ratio, the exact weight of the sample is not as important since relative ratios are independent of the weight used. This type of analysis can be helpful if MOF linkers or nodes have been metalated (e.g., for catalysis) with a metal that is different from that already present in the MOF node. ICP-OES/MS can also be used to detect metal (or linker) leaching from an MOF into solution, for example, during a catalytic reaction, or when testing analyte uptake into an MOF from solution. It should be noted that hydrofluoric acid (HF) can also be used to digest MOF samples prior to ICP analysis, although it is rarely necessary and caution should be taken when handling HF. The elements that can be measured using ICP are listed here.¹⁰⁹

NMR Spectroscopy. NMR spectroscopy can be used to determine MOF purity, linker ratios, and leftover modulator, as

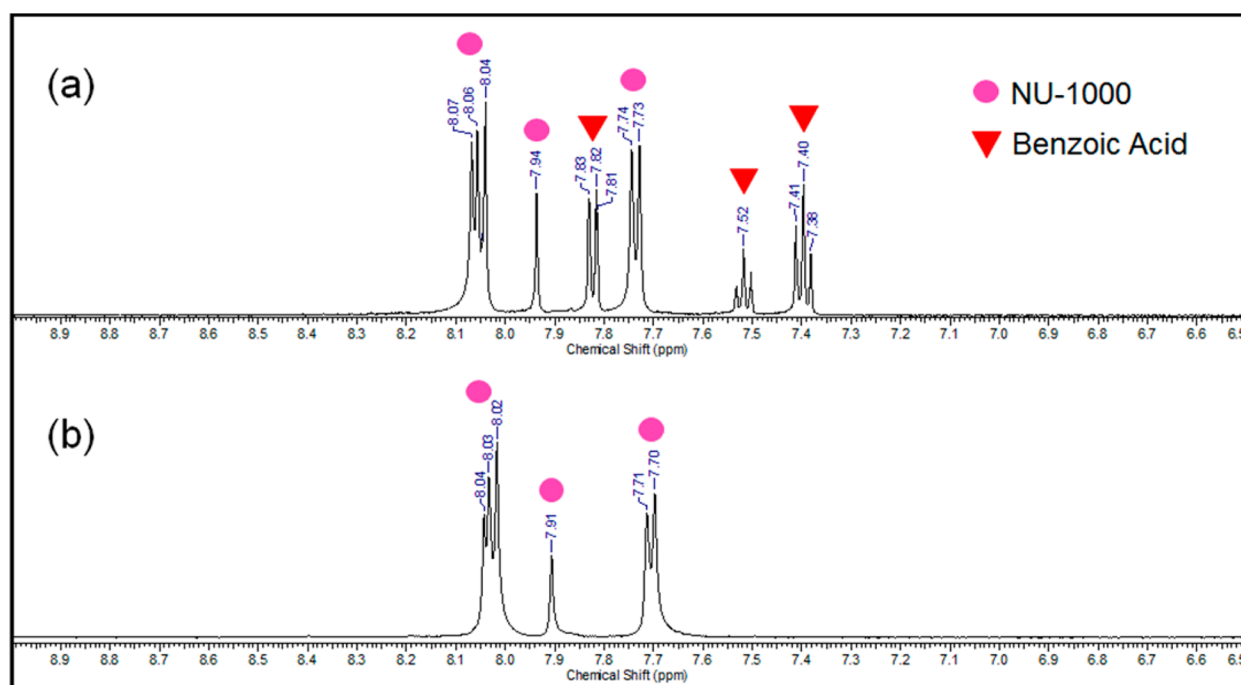


Figure 9. NMR spectrum of (a) NU-1000 prior to removal of excess benzoic acid modulator from the Zr_6 -node and (b) NU-1000 after removal of benzoic acid.

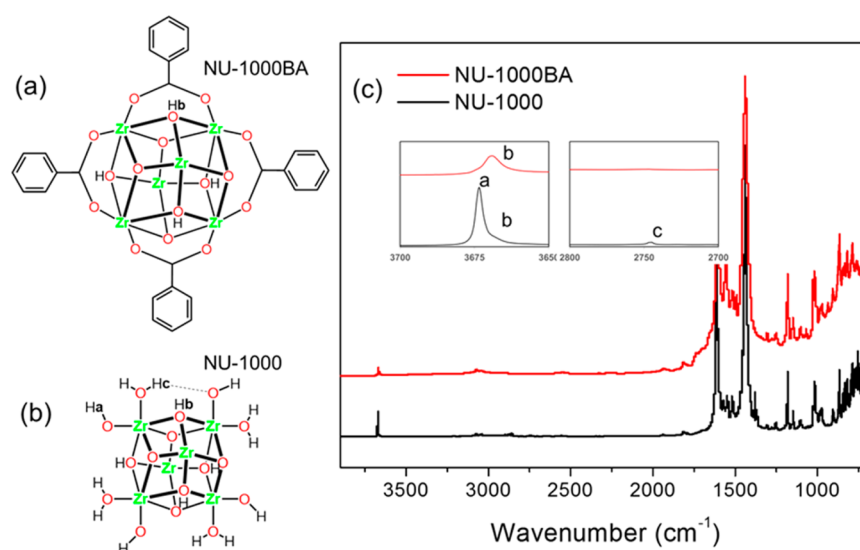


Figure 10. Zr_6 node of NU-1000 before (a) and after (b) removal of excess benzoic acid modulator; (c) diffuse reflectance infrared Fourier transform spectrum of NU-1000 before and after removal benzoic acid. Insets show the regions from 3700 to 3650 cm^{-1} and 2800 cm^{-1} to 2700 cm^{-1} , respectively, with peaks labeled according to proton labels on the nodes.

well as the absence of solvent after activation. Given that many MOFs are not soluble in conventional NMR solvents, MOF digestion is often required before a spectrum can be obtained. The most common¹¹⁰ method of MOF digestion for NMR purposes involves adding anywhere from 5–10 drops of D_2SO_4 to 1–2 mg of MOF and sonicating the mixture until the sample is well dispersed in the acid. Approximately 0.5 mL of $DMSO-d_6$ can then be added to dilute and fully dissolve the mixture (sonication and heat may be required after the addition of $DMSO-d_6$ to dissolve the sample). Similar to ICP-OES analysis, the sample must be fully dissolved for the data analysis to be meaningful. If the procedure described above does not result in a fully dissolved sample, less MOF and/or more acid and heat

can be used as required. Alternatively 0.1 M NaOD (made by diluting 40 wt % NaOD with D_2O) can be used in place of D_2SO_4 . Figure 9 shows NMR spectra of NU-1000 before and after removal of excess benzoic acid modulator from the Zr_6 -nodes of the framework. This is one example where NMR analysis is helpful to determine the successful “purification” of a sample. Hydrofluoric acid (HF) diluted with D_2O can also be used to digest MOF samples¹⁰⁶ for NMR analysis; however, it is rarely required, and caution should be taken when handling HF.

Solid-state NMR (SS-NMR) spectroscopy can also be used for MOF characterization,^{111,112} particularly when probing the local chemical environment inside an MOF.^{113–116} This

technique can be extremely useful for characterizing the identity or chemical state of a specific functional group inside an MOF, such as elucidating the oxidation state of a phosphorus atom inside an MOF.¹¹⁷ SS-NMR can also be used to gain understanding about the supramolecular interactions and kinetics of small molecule docking inside an MOF.¹¹⁸ The SS-NMR technique that should be used such as cross-polarization (CP) or direct polarization (DP) magic angle spinning (MAS), for example, is dependent on the MOF and the information that is to be extracted from the experiment.¹¹⁹ Dynamic nuclear polarization (DNP) enhanced SS-NMR spectroscopy can also be used for MOF analysis to increase sensitivity and decrease data collection time.¹²⁰ It is important to note that a significant sample size (about 1 cm³) should be used to obtain meaningful SS-NMR data.

Diffuse Reflectance Infrared Fourier Transform Spectroscopy. DRIFTS can be used to probe the presence or absence of IR active functional groups in an MOF. With a specialized sample holder, more complex information can be obtained such as the behavior of an MOF in the presence of specific gases (e.g., CO, CO₂) or at varying temperatures. DRIFTS can simply be performed on powdered samples of an MOF, or the samples can be diluted in a matrix such as potassium bromide (KBr). One way to dilute the sample involves physically mixing the MOF with KBr using a mortar and pestle. Caution should be taken when grinding the sample as some MOFs are not stable to mechanical stress. When looking for IR stretches ranging from 3500–3800 cm⁻¹, it is important that water is not present in the sample or in the KBr matrix as the large and broad –OH stretches from water can drown out peaks in this region. To ensure the absence of water, the sample should be activated prior to DRIFTS analysis or alternatively the sample can be heated in the spectrometer if a variable temperature sample holder is available. Figure 10 shows a DRIFTS spectrum for NU-1000 before and after removal of excess benzoic acid from the Zr₆ nodes of the framework. After removal of benzoic acid, the sharp –OH stretch at 3670 cm⁻¹ corresponding to the terminal –OH groups on the Zr₆ node (Table 1) becomes visible, as well as a small peak at 2740 cm⁻¹ corresponding to hydrogen bonding water on the node. This is one example where DRIFTS analysis can be used (complementary to other techniques) to determine the successful “purification” of a sample.

Single Crystal X-ray Diffraction. The best method to unambiguously determine the structure of an MOF is through the use of single crystal XRD. As is the case with XRD on any material, the major limiting factor to collecting reliable data is governed by the size and quality of the crystal grown. In general, crystals should be greater than 5–10 μm in size, which can be challenging to obtain for some MOF families. Unfortunately, there is not a general procedure that can be used to obtain larger MOF crystals, and typically reaction conditions such as temperature, time, modulator, and reagent concentration must be systematically varied to determine the optimized conditions for increasing crystal size in each individual case. It is also important to note that even if a large crystal is obtained, other characterization methods (described above) should be used to ensure that the crystal matches the structure of the bulk MOF material. MOF crystallography is often challenging due to large amounts of disordered solvent in the pores of these materials, and therefore optimizing the temperature for data collection is very important. A common method used for eliminating disordered

solvent in a structure is SQUEEZE, although a number of criteria must be followed to use this method correctly.¹²¹ It is often assumed that collecting data at low temperature is the best practice for MOF crystals, but this is not necessarily the case. Higher temperatures can sometimes be used to give better diffraction data since disordered solvent will freely tumble (becoming highly disordered) and not contribute much to diffraction peaks.⁹ Unactivated MOFs (synthesis-solvent-filled MOFs) typically diffract better than activated versions. The major challenges with MOF structure refinement are related to symmetry mismatches between the lattice of the nodes (usually higher symmetry) compared to the lattice of the organic linkers (usually lower symmetry). As a result, MOF structures that contain linkers with higher symmetry are often less challenging to refine. Interpenetration of MOFs can also make crystallographic characterization of MOFs very challenging,¹²² although interpenetrated MOFs often show better diffraction since they are more dense and have less void space.

CONCLUSIONS

In summary, we have outlined our view of the current best practices for the most common techniques and methods used to synthesize, activate, and characterize MOFs. We hope that this tutorial article will be particularly helpful for scientists who are new to the field of MOF chemistry, in addition to stimulating productive discussion among seasoned researchers. To the extent that best practices can be widely adopted, meaningful comparisons between materials and data from different experimental laboratories, as well as meaningful comparisons between experimentally measured and computationally derived data, will be facilitated. MOF chemistry is a continually developing and expanding field, and with new developments come new techniques and the fine-tuning and evolution of best practices.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.chemmater.6b02626.

- Summary of and photographs from the videos (PDF)
- Video 1: Preparation of UiO-67 using benzoic acid or HCl as a modulator (MP4)
- Video 2: Synthesis of NU-1000 (MP4)
- Video 3: Solvent exchange/product isolation (MP4)
- Video 4: Supercritical CO₂ activation (MP4)
- Video 5: ICP sample digestion (MP4)

AUTHOR INFORMATION

Notes

The authors declare no competing financial interest.

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